

RESEARCH REPORT

Automatic License Plate Readers: Legal Status and Policy Recommendations for Law Enforcement Use

Summary: The proliferation of ALPR technology raises serious civil rights and civil liberties concerns. Courts, lawmakers, and technology vendors must take action.



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Americans drive. According to one survey, 83 percent of U.S. adults drive a car at least several times a week. ^[1] In jurisdictions with limited or no public transportation, driving may even rival cell phone use as a modern necessity.

Cars connect people with work, love, school, prayer, and protest.

They also leave a data trail. Historically, it would have been virtually impossible for law enforcement to routinely surveil all drivers. However, with the growing use of automatic license plate readers

(ALPRs), police can now receive alerts about a car's movements in real time and review past movements at the touch of a button. ALPRs could prove valuable in police investigations and for non-law enforcement uses like helping government agencies to reduce traffic and curb environmental pollution. But legal and policy developments have failed to adequately address the risks posed by this highly invasive technology. [\[2\]](#)

Recent events crystalize ongoing concerns. With Black Lives Matter demonstrations taking place across the United States in the wake of the George Floyd and Breonna Taylor murders, law enforcement agencies large and small are deploying their expansive surveillance arsenals to monitor protesters. For many agencies, those surveillance tools include ALPRs, which have heightened relevance in localities where people must drive to protests, or if protests themselves are occurring by car, as is increasingly happening during the ongoing Covid-19 pandemic. [\[3\]](#)

The pandemic adds an additional dimension for consideration, as states look for creative ways to control the virus's spread. With car travel expected to increase as states begin slowly loosening restrictions, ALPRs may play a larger role in law enforcement. [\[4\]](#)

States such as Rhode Island have already directed law enforcement to look for New York license plates in order to identify people who should be directed to self-quarantine. [\[5\]](#) Law enforcement agencies may look to automate this process by using ALPR devices to alert officers any time an out-of-state license crosses into their localities.

This white paper explains how ALPR technology works, focusing on its use by law enforcement agencies. It then analyzes both the legal and policy landscapes, including how courts have ruled on the use of ALPRs, and how they can be expected to rule in the future. Next, it outlines a series of concerns, ranging from high error rates to the impact on civil liberties and civil rights. Finally, it concludes with a set of recommenda-

tions for law enforcement, lawmakers, and technology vendors to enhance transparency and accountability and

mitigate the impact of this technology on individuals' civil liberties and civil rights.

How Do Automatic License Plate Readers Work?

Automatic license plate readers use a combination of cameras and computer software to indiscriminately scan the license plates of every car passing by. The readers, which can be mounted on stationary poles, moving police cruisers, and even handheld devices, log the time and date of each scan, the vehicle's GPS coordinates, and pictures of the car. Some versions can also snap pictures of a vehicle's occupants and create unique vehicle IDs. ^[6] The devices send the data to ALPR software, which can compare each plate against a designated "hot list." Such lists can include stolen cars and cars associated with AMBER Alerts for abducted children. ^[7] They can also reference vehicles

that are listed in local and federal databases for reasons that may include unpaid parking tickets or inclusion in a gang database. ^[8] These queries happen automatically, though officers can also query plates manually. ^[9]

In addition to checking data in real time, many cities and agencies retain plate information for future use, sometimes indefinitely. ^[10] This data can be used to plot a particular vehicle's various locations or to identify all the cars at a given location, and it can even be analyzed to predict routes and future locations of a vehicle or set of vehicles. ^[11] These tools may cost little or nothing for police, often because the drivers themselves shoulder the cost of the technology through a fee charged on top of traffic ticket costs. ^[12] Notably, drivers in some jurisdictions can be jailed for

failure to pay the private company's fee. ^[13]

Law enforcement use of ALPRs is rapidly expanding, with tens of thousands of readers in use throughout the United States; one survey indicates that in 2016 and 2017 alone, 173 law enforcement agencies collectively scanned 2.5 billion license plates. ^[14]

According to the latest available numbers from the Department of Justice's Bureau of Justice Statistics, 93 percent of police departments in cities with populations of 1 million or more use their own ALPR systems, some of which can scan nearly 2,000 license plates per minute. ^[15] In cities with populations of 100,000 or more, 75 percent of police departments use ALPR systems.

^[16] In some of the largest U.S. cities, millions of license plates are scanned over the course of a year. ^[17]

According to a 2020 California state auditor report, the Los Angeles Police Department (LAPD) alone has accumulated more than 320 million license

plate scans, and the Sacramento Police Department recorded up to 1.7 million scans in just one week. ^[18] Despite this expansive data collection effort, many departments have not developed a policy to govern the use of ALPR technology, or provided privacy protections. While states such as California and Nebraska have passed laws requiring their departments to establish ALPR policies, not all departments have complied. ^[19]

Law enforcement use of ALPR data is not limited to reads captured by departments' own devices; many departments have contracts with vendors that grant them access to private databases containing scans from private ALPRs and from other local and federal law enforcement agencies. For example, Vigilant Solutions (owned by Motorola Solutions), a leading provider of ALPR data to police based in Livermore, California, sells access to its database of more than 5 billion license plate scans collected across the country, including 1.5 billion reads provided by law

enforcement agencies. This process creates a revolving door of license plate scans from law enforcement to Vigilant Solutions back to law enforcement agencies. ^[20]

Moreover, access to ALPR tools and data is not limited to law enforcement. For example, government agencies use license plate readers to automate toll collection and for pollution research; businesses analyze ALPR location data when assessing loan applications to help verify an applicant's listed home address or to detect commercial use of

vehicles when analyzing insurance claims; and private individuals and neighborhood associations can buy ALPRs for home and neighborhood security purposes. ^[21] These private actors can maintain their own hot lists of flagged license plate numbers and can share any data they collect with law enforcement at their discretion. ^[22] Similarly, public agencies that collect and store ALPR data for non-law enforcement purposes may hold onto a dataset that proves alluring for police departments.

What Does the Law Say?

The U.S. Constitution's Fourth Amendment protects people from unreasonable searches and seizures. ^[23] According to the U.S. Supreme Court, the Amendment's purpose "is to safeguard the privacy and security of individuals against arbitrary invasions by government officials." ^[24] Until the late

1960s, the Supreme Court ruled that Fourth Amendment protections only applied to searches and seizures of tangible property. ^[25] But in 1967, the Court expanded Fourth Amendment protections, holding in *Katz v. U.S.* (1967) that "the Fourth Amendment protects people, not places." ^[26] Specifically, the government was now prohibited from intruding upon a person's "reasonable expectation of privacy." In other words,

if an individual seeks to keep something private, and that expectation of privacy is “one that society is prepared to recognize as reasonable,” the Fourth Amendment is triggered, and the government generally must obtain a warrant supported by probable cause before conducting a search. ^[27] This approach seeks to protect the “privacies of life” from “arbitrary power,” and to “place obstacles in the way of a too permeating police surveillance.” ^[28]

By contrast, the Court has not required a warrant or other heightened standard for police officers to take pictures of individual license plates and compare them against a law enforcement database. Its reasoning has been twofold. First, due to “the pervasive regulation of vehicles capable of traveling on the public highways,” there is no expectation of privacy in the content of license plates. ^[29] Second, longstanding precedent holds that drivers on public roads cannot expect their movements to be kept private from the police since they could be observed by any member

of the public (though, as discussed below, this presumption is beginning to shift). In keeping with these doctrines, courts have regularly held that law enforcement officers may, at their discretion and without any suspicion of criminal activity, perform at least an initial check of a license plate against a law enforcement database. ^[30]

Even so, there have long been hints that the tracking of vehicles’ movements could, under some circumstances, trigger Fourth Amendment concerns. As far back as 1979, the Supreme Court declared that “an individual operating or traveling in an automobile does not lose all reasonable expectation of privacy simply because the automobile and its use are subject to government regulation.” ^[31] Similarly, when the Court analyzed the use of beeper technology in the 1980s, it distinguished limited monitoring from “twenty-four hour surveillance of any citizen in the country,” reserving the question of whether such “dragnet type law enforcement prac-

tices” merit the application of different constitutional principles. ^[32]

More recently, the Court’s application of the Fourth Amendment has evolved significantly in response to technological “innovations in surveillance tools.” ^[33]

In *Kyllo v. U.S.* (2001), for instance, the Supreme Court held that police need a warrant before they can use a thermal imager to detect heat coming from a garage. By doing so, the Court rejected a return to a “mechanical interpretation” of the Fourth Amendment, under which the Constitution would have protected only against physical intrusions into a person’s private space, holding instead that it was necessary to ensure that people were not left “at the mercy of advancing technology.” ^[34] Over time, the Court has ruled that law enforcement must obtain a warrant before searching a suspect’s cell phone during an arrest (even though it had previously allowed warrantless searches incident to arrest), before installing a GPS tracker on an automobile for long-term monitoring (despite precedent suggest-

ing that vehicular movements are not private), and before obtaining historical cell-site location information revealing an individual’s daily movements (although third-party information can normally be obtained without a warrant). ^[35]

The reasoning in these cases is instructive. Take *U.S. v. Jones* (2012), in which the Supreme Court held that the police need a warrant in order to install a GPS tracking device on a car and use it for extended surveillance. In her concurrence, Justice Sonia Sotomayor observed that inexpensive location tracking “makes available at a relatively low cost such a substantial quantum of intimate information about any person whom the Government, in its unfettered discretion, chooses to track” that it “may ‘alter the relationship between citizen and government in a way that is inimical to democratic society.’” ^[36]

Similar themes run through the Court’s decision in *Carpenter v. U.S.* (2018), which holds that police must get a warrant before they can obtain historical in-

formation from cell phone providers about the location of individuals' mobile phones (known as cell-site location information, or CSLI). ^[37] The Court observed that this information could be used to track the minutiae of people's daily lives. It reasoned that the "depth, breadth, and comprehensive reach" of this data, along with "the inescapable and automatic nature of its collection" by virtue of simply carrying a cell phone, necessitate a warrant supported by probable cause. ^[38]

While the *Carpenter* decision narrowly addresses the use of historical CSLI, it provides an important framework for analyzing reasonable expectations of privacy in the digital age. Location tracking via ALPR databases raises many of the same concerns outlined in *Carpenter*; an application of its framework should lead courts to conclude that police must first obtain a warrant before searching historical location information from ALPR databases.

Specifically, first, *Carpenter* instructs courts to consider the capacity of a technology to enable ongoing surveillance that would have been unimaginable before the digital age. ^[39] Just as with CSLI, automatic license plate readers enable data collection that is "detailed, encyclopedic, and effortlessly compiled." ^[40] A person's phone is constantly creating records simply by being powered on and connecting to the network. Similarly, ALPRs automatically collect information about every car that passes within their range. But while a person might turn off their cell phone while they travel, it may be almost impossible to avoid traveling some roads without exposing one's vehicle to ALPRs.

Second, the *Carpenter* Court considered the extent to which data collection is indiscriminate, targeting not only people under investigation but a much broader segment of the population.

^[41] While ALPR scans provide a different level of pinpoint accuracy than CSLI, they also indiscriminately collect

data about every car that passes by a license plate reader, regardless of the driver's connection to criminal activity. In fact, the vast majority of scans capture information about drivers who are not suspected of any wrongdoing. ^[42] The only limitations on this ongoing surveillance of all cars traveling a public road are the number of ALPRs and the data retention policies maintained by police or third-party vendors.

Third, the Court considered the extent to which the long-term CSLI retention allowed officers to effectively create a time machine of a person's movements. Just as with historical CSLI, the long-term retention of plate data allows the police to retroactively track every location where a particular car was tagged by an ALPR device. ^[43] To be sure, the current scope of ALPR devices does not match the scope of cell phone towers blanketing the country, which makes a direct comparison difficult.

Nonetheless, the current adoption rate of ALPRs suggests that this technology will continue to expand its coverage ar-

eas. In fact, the *Carpenter* Court ruled that lower courts "must take account of more sophisticated systems that are already in use or in development." ^[44] ALPR technology is expanding at a rapid rate, with growing databases containing billions of license plate scans, and with governmental and private ALPR devices capturing larger swaths of cities. Courts should consider this foreseeable future when confronted with nascent uses of ALPR that appear smaller in scale.

Finally, the Court ruled that an interpretation of the Fourth Amendment called the third-party doctrine is inapplicable to historical CSLI. ^[45] Under the third-party doctrine, individuals do not have a reasonable expectation of privacy in information they are deemed to have voluntarily handed over to third parties.

^[46] The *Carpenter* Court found that while this doctrine is appropriate for limited disclosures such as bank records or a log of dialed telephone numbers, it should not apply to CSLI data, which can provide a "chronicle of

a person's physical presence compiled every day, every moment, over several years." ^[47] Historical ALPR data similarly chronicles the movements of all vehicles, regardless of the registered owner's connection to a suspected crime.

The *Carpenter* Court also reasoned that individuals do not truly voluntarily share their location data with wireless carriers; instead, the data is automatically collected simply by possessing a cell phone — a device the Court described as "indispensable to participation in modern society" — and by connecting to a mobile network. ^[48] Similarly, a majority of Americans rely on driving in order to fully participate in society, and their movements are logged by ALPRs by virtue of simply driving and parking on public roads. Just as the only way to avoid generating CSLI would be to turn off a mobile device, the only way to avoid ALPR data collection would be to give up driving altogether or to keep a vehicle away from the range of a license plate reader — an impossible task in

many places. ^[49] *Carpenter* thus suggests that the third-party doctrine is equally inapplicable to historical location data collected by ALPR devices.

Although the Supreme Court has not yet addressed whether police access to historical ALPR data requires a warrant, appeals courts have begun hearing challenges to warrantless ALPR database searches. However, courts appear reluctant to embrace a bright-line rule that extends *Carpenter* to ALPR searches. The result has been a series of one-off decisions that seek to avoid direct engagement with the foreseeable proliferation of ALPR data.

For example, in *U.S. v. Yang* (2020), the Ninth Circuit ruled that the defendant did not have standing to challenge government queries of a private ALPR database for records of his rental car travels when he kept the vehicle past the contract due date in violation of company policy. ^[50] This ruling now compels defendants in the Ninth Circuit to prove that they had a sufficiently

close relationship with the property that was searched before the court will address their Fourth Amendment rights.

And in *Commonwealth v. McCarthy* (2020), the Massachusetts Supreme Judicial Court ruled that while widespread use of ALPR devices *can* implicate a person's Fourth Amendment privacy interest in the whole of their movements, the limited surveillance undertaken in that case did not violate the defendant's reasonable expectation of privacy.^[51] The case involved police officers' use of ALPR hot list notifications to track the defendant's movements as he traveled across two bridges over the course of two months.

^[52] The court applied what is commonly referred to as the "mosaic theory," where the long-term surveillance of a person's public movements triggers a privacy interest that could be absent with only limited or isolated monitoring.

^[53] The court acknowledged that "with enough cameras in enough locations, the historic location data from an ALPR system in Massachusetts would invade

a reasonable expectation of privacy and constitute a search for constitutional purposes."^[54] Four ALPR devices on

two bridges did not, however, rise to this level, according to the ruling.^[55]

Lower courts will continue to face the dilemma of how to rule on the specific use of ALPRs in a given case while taking into account the Supreme Court's admonition that courts must consider the logical evolution of these systems of surveillance.^[56]

Separate from Fourth Amendment considerations, courts have also considered how ALPR technology may violate privacy protections under state law. For example, the Virginia Supreme Court is currently hearing an appeal seeking to reopen the substantive issue of whether the Fairfax County Police Department's use of an ALPR system to passively track the movements of cars that were not on a hot list violates the state's Government Data Collection and Dissemination Practices Act.^[57] This act requires, among other things, that

information not be collected unless the need for it has been clearly established ahead of its collection — a standard that indiscriminate collection of ALPR data cannot meet. ^[58] If the trial court's ruling is upheld, the Fairfax County Police will be required to purge ALPR data that is not linked to a criminal investigation and to stop using ALPRs to passively collect data on people who are not suspected of criminal activity. ^[59]

ALPRs are relevant to more than privacy. Courts have also considered whether an ALPR hit provides sufficient justification for a police officer to stop a car. In *Kansas v. Glover* (2020), the Supreme Court ruled that a license plate search indicating that a car's registered owner has had his or her license revoked gives police reasonable suspicion to perform a traffic stop in the absence of information suggesting that someone other than the owner is driving the vehicle. ^[60] Several state courts reached the same conclusion. ^[61]

An ALPR hit is not always a sufficient basis for a stop, however. ^[62] In 2014, the U.S. Court of Appeals for the Ninth Circuit considered an erroneous ALPR alert that led to a traffic stop in which a woman was detained and held at gunpoint. ^[63] Unlike in *Glover*, where an officer manually searched a license plate number and confirmed that the truck he observed matched the vehicle in the database, in *Green v. City and County of San Francisco* (2014), an ALPR device mounted on an officer's cruiser malfunctioned and returned a hit for a different vehicle and license plate number than the plaintiff's car. ^[64] An officer radioed in a description of the plaintiff's vehicle, along with the incorrect license plate number picked up by the ALPR device. ^[65] A second officer identified the plaintiff's car, but did not attempt to confirm whether the radioed license plate number matched the plates on the plaintiff's car. The Ninth Circuit ruled that the case could proceed on the question of whether the second officer should have taken addi-

tional steps to independently confirm whether the ALPR device had identified the right car and license plate number before initiating a traffic stop. ^[66] The *Green* decision, which analyzed an

ALPR system that “frequently” makes mistakes, may suggest that there are situations in which reliance on an ALPR hit remains insufficient to justify a traffic stop.

Policy Concerns

In light of the wide saturation of license plate readers, it is critical that the use of these devices be accurate, bias-free, and protective of established legal values and constitutional rights.

Unfortunately, publicly available information suggests that this is not the case. This may explain why at least 16 states have passed laws regulating the use of ALPRs or the use of data collected by the devices. ^[67] Some prohibit the use of ALPRs except for limited public safety purposes, whereas others establish controls governing their use, including mandatory privacy policies, limits on data retention, express limits on the types of investigations in which they can be used, and mandatory audits. ^[68] These regulations highlight

many of the concerns around ALPRs listed below and predict many of the recommendations that follow.

High error rates: Errors can arise in at least two ways — inaccurate hot lists and inaccurate reads. If hot lists are not updated, an individual may be pulled over when, for instance, the system incorrectly indicates that a license is suspended when it has actually been reinstated. Inaccurate reads are surprisingly common as well: one randomized control trial in Vallejo, California, found that 37 percent of all ALPR “hits” from fixed readers (such as those attached to a street light) and 35 percent from mobile ALPRs were misreads — an astonishingly high error rate. ^[69] In several high-profile incidents, drivers have been

pulled over because a reader read the numbers on their license plates wrong and erroneously tagged the vehicles as stolen. ^[70] In one instance, a Colorado woman and several children were detained and handcuffed face-down on the ground after an ALPR mistook their SUV for a stolen motorcycle from a different state. ^[71] In another, the chair of Oakland's Privacy Advisory Commission was mistakenly stopped and detained at gunpoint after his rental car's license plate triggered an out-of-date hit signaling to police officers that the car had been stolen. ^[72] Even in cases where a vehicle is accurately flagged, it may not convey accurate information about an individual. A car can be shared among family members, among friends, or as part of a carshare; this reality may place low-income individuals, who are more likely to share cars, at greater risk of misidentification.

To be sure, license plate readers have had some high-profile successes: a man accused of stabbing several people af-

ter breaking into a rabbi's home during a Hanukkah celebration was found in part due to an alert from an ALPR device; a Tennessee girl abducted by her noncustodial father was recovered when a license plate camera spotted his car; and police were able to use information from a license plate reader to help halt a string of random shootings on highways in Kansas City, Missouri.

^[73] Despite these anecdotal successes, there has not been a thorough assessment of the tool's value. Any such assessment would require consideration of the ALPR's additional costs and benefits described here.

Privacy and data security concerns:

An extremely small percentage of cars scanned by ALPRs — generally far below 1 percent — are connected to any crime or wrongdoing. ^[74] For example, an audit found that 99.9 percent of the ALPR images stored by the LAPD are for vehicles not on a hot list at the time a license plate was scanned. ^[75] Nevertheless, many jurisdictions keep the scans "just in

case,” storing the data for anywhere from 90 days to two years or even indefinitely. ^[76] These scans, over time, can reveal individuals’ movements and help create detailed pictures of their private lives. ^[77]

In addition to information generated by ALPRs, police officers can also add to and store sensitive information in the databases housing license plate scans through open text fields and hot lists available in the user interface. For example, the California state auditor found that law enforcement can input information including personal information such as names, addresses, dates of birth, and physical descriptions, and they can also store criminal justice information such as criminal charges and warrant information. ^[78] License plate readers have also been known to capture private information, such as shots of children exiting a car in the driveway of a home or activity inside an open garage — information that surely should not be retained. ^[79] This is information

that goes far beyond the legitimate need to find stolen cars or vehicles linked to AMBER Alerts. The ongoing storage of this wide array of sensitive information also raises security concerns, as this information can be vulnerable to data breaches and hacking. The data security applied to ALPR data may not be commensurate with the sensitivity of the data being held.

Data sharing concerns: Many vendors allow their law enforcement clients to share and receive ALPR data from other law enforcement agencies. For example, through Vigilant Solutions’ Law Enforcement Archival Reporting Network (LEARN), police departments can elect to automatically share their collection of license plate reads with outside law enforcement partners that are also part of the network. These data sharing arrangements are not always made public or adequately tracked by police departments, which can result in impermissible or unaccountable sharing. An ACLU investigation found that

more than 80 local police departments had set up their LEARN settings to share ALPR data with U.S. Immigrations and Customs Enforcement (ICE), even though the practice may violate local privacy laws or sanctuary policies. ^[80] Local laws and policies will have limited effect if they do not address automated data sharing or if law enforcement cannot effectively control data flows in and out of their departments. In one instance, the California state auditor found that despite efforts to limit data sharing with ICE, confusing vendor settings had left three different ICE agencies with access to ALPR data from Marin County Sheriff's Office, frustrating compliance with a California law that places controls on local police cooperation with immigration authorities. ^[81] Customs and Border Protection also receives ALPR data from commercial vendors, including information from across the United States, "outside of the border

zone in which CBP activities take place." ^[82]

With public agencies seeking to collect ALPR data for uses such as toll collection or environmental analysis, there are concerns that the information being collected may be intentionally or unintentionally shared with law enforcement agencies. Without policies governing the type of data that is collected, stored, and shared, these government data sets may create a frictionless data sharing opportunity that frustrates attempts to limit law enforcement access to ALPR data. For example, in San Diego, law enforcement officers regularly access smart streetlight footage — in some cases, to surveil Black Lives Matter protests — even though the streetlights project was originally intended to assist city planners and app developers. ^[83]

As more companies sell ALPRs to homeowners, additional data sharing concerns emerge. For example, this trend allows police officers to expand the reach of their surveillance systems

by providing them with access to private device feeds that may be outside the scope of law enforcement policies governing their own equipment (if any exist at all). When police officers solicit data from private ALPR systems, the decision to share information is up to the individual homeowner or the private company providing the service. While companies may maintain privacy policies that explain the situations in which they share information with law enforcement, the policy only covers the company and the person who purchased the devices. ^[84] The registered owners of vehicles tracked and logged by these private devices will not receive notice or an opportunity to object to data sharing arrangements between police and private individuals.

Layering ALPR data with other surveillance systems:

License plate readers can be used alongside other kinds of technologies to facilitate even more widespread surveillance. The New York City Police Department (NYPD) Domain Awareness System,

for instance, can track the movements of cars and people using its 20,000 security cameras and license plate readers (along with face and object detection technology). ^[85]

Several police departments, including those in Chicago, Detroit, and Memphis, incorporate ALPR technology into Real Time Crime Centers (RTCCs) that combine license plate data with surveillance footage from thousands of police, school, and traffic cameras, gunshot detection systems, and social media monitoring.

^[86] The expansive and ongoing monitoring that is facilitated through these integrated systems may be incompatible with constitutional freedoms, including the right to free assembly and the right to privacy. These burdens hinder the collective organizing necessary to hold the government accountable to the will of the people.

Lack of transparency and access

controls: While ALPRs are increasingly ubiquitous, many police departments do not actively maintain use

policies, and there are insufficient controls to protect against misuse.

[\[87\]](#) A 2016 investigation by the Associated Press found that police officers across the country abuse confidential databases to spy on love interests, journalists, business associates, and others. [\[88\]](#) For example, the investigation reported officers stalking ex-girlfriends, looking up the addresses of crushes, and in one case, running searches on a journalist who wrote a series of stories critical of the department. ALPR databases could easily be put to similar use. [\[89\]](#)

Police departments exacerbate this problem when they fail to implement access and monitoring safeguards to ensure that ALPR data is only accessed on a “need to know” basis, and that access is appropriately logged and monitored to protect against misuse. Instead, some departments automatically install ALPR software on every computer assigned to staff, even when their position does not require access to this kind of information.

[\[90\]](#) There is a further lack of transparency in many jurisdictions where vendor contracts prohibit police departments from disclosing their use of surveillance systems to the public.

[\[91\]](#)

Disparate impact concerns: ALPRs can also be deployed to target communities of color or other vulnerable populations. The NYPD has used license plate readers as part of its widespread surveillance of Muslim communities in the New York and New Jersey area. [\[92\]](#) And an investigation of license plate readers in Oakland, California, found that they were located predominantly in Black and Latino neighborhoods, despite the fact that automobile crimes and offenses predominantly occurred elsewhere. [\[93\]](#) Even the placement of ALPRs in “high crime” neighborhoods will likely reflect a history of biased and selective enforcement that has already led to the over-policing of communities of color. The guise of

neutral surveillance will only reinforce these practices and maintain the attendant potential for deadly police encounters.

Some police departments also incorporate ALPR data into gang databases, which allows officers to track vehicles associated with suspected gang members. ^[94] These gang databases are notoriously unreliable, as they rely on vague and often contradictory criteria for inclusion. ^[95] Gang databases, which contain tens of thousands of names, are almost overwhelmingly comprised of individuals of color, and people frequently have no opportunity to challenge their inclusion. ^[96] The LAPD suspended use of California's statewide gang database after announcing audits and investigations in response to allegations that police falsified records and listed innocent people as gang members. ^[97] Meanwhile, an audit by Chicago's Office of Inspector General found that the city's gang database contained incomplete and con-

flicting data, with some entries raising serious concerns about how officers "perceive and treat the people with whom they interact." ^[98]

In the wake of nationwide protests that followed the police killings of George Floyd and Breonna Taylor, public attention has increasingly focused on the ongoing instances of police brutality and racial bias in policing. However, there is a risk that police departments and legislators may incorrectly propose surveillance as a neutral alternative.

Surveillance that disproportionately targets communities of color carries a distinct and cognizable equal protection harm: branding them with a badge of inferiority. As one appellate court wrote, "Our nation's history teaches the uncomfortable lesson that those not on discrimination's receiving end can all too easily gloss over the 'badge of inferiority' inflicted by unequal treatment itself. Closing our eyes to the real and ascertainable harms of discrimination inevitably leads to morning-after regret."

^[99]

Impact on protected First

Amendment rights: Law enforcement agencies have a history of misusing license plate surveillance to monitor First Amendment-protected activity. During the 2008 presidential election, the Virginia State Police recorded the license plate numbers of attendees at political rallies for Barack Obama and Sarah Palin — and subsequently at President Obama's inauguration — and kept the data for more than three years until it was purged following an opinion from the Virginia Attorney General warning that ongoing retention would violate the state's Government Data Collection and Dissemination Practices Act.

^[100] Similarly, police in Denver spied on anti-logging activists and shared license plate information with the FBI's Joint Terrorism Task Force when the activists held a training on nonviolence. ^[101]

Such surveillance — whether it involves the locations of multiple cars that appear together in the same place, or of a

single car at places like a mosque, synagogue, or rally — has a chilling effect on Americans' First Amendment rights to freedoms of association, religion, and speech. ^[102] An investigation into an NYPD program that monitored mosque visitors' license plates found that this surveillance “chilled constitutionally protected rights — curtailing religious practice, censoring speech and stunting political organizing.” ^[103] The International Association of Chiefs of Police has noted that ALPRs can cause people to “become more cautious in the exercise of their protected rights of expression, protest, association, and political participation because they consider themselves under constant surveillance.” ^[104] And there is always the specter of more flagrant abuse, such as putting a political opponent's license plate on a hot list and using it to keep track of that person's whereabouts.

^[105]

Recommendations

In light of these concerns, the need for a multifaceted response to the proliferation of ALPR use is overdue. This section contains a number of recommendations for policymakers, law enforcement agencies, and technology vendors. These are intended as starting points, as the circumstances and implementation of ALPR use reforms will vary by jurisdiction.

Adopt retention limits and require warrants for searching historical

data: Plates that are scanned and do not match a hot list alert should be promptly discarded. Using ALPRs to record the movements of all vehicles in a municipality goes far beyond the limited information that is revealed by an isolated capture of license plate data. It also goes beyond what an ordinary person could observe on a public road. Alternatively, if plates are retained, the retention period should be

as brief as possible — on the scale of days, not months. These limits could be imposed both by departmental policies and by state law. If municipalities elect to permit retention of license plate data to enable historical searches, such searches should require a warrant supported by probable cause absent emergency situations. ALPR databases collect expansive and sensitive accounts of people's movements regardless of whether they are suspected of criminal activity in a manner that is not available through traditional surveillance. Aside from ceasing to drive altogether, it is exceedingly difficult to avoid this type of surveillance — a condition that is likely to become more pronounced as ALPR technology continues to expand.

Institute a two-step scanning

process: When it comes to officers individually running plates, police departments should tailor their scanning processes so that the first pass

through a database will not yield protected personal information, instead revealing only registration information and presence on a hot list. Only if that first inquiry reveals a “basis for further police action” should the officer be permitted to proceed to a second step, which would “allow access to the ‘personal information’ of the registered owner, including name, address, social security number, and if available, criminal record.” ^[106] This simple process can help deter police use of these systems for purposes other than law enforcement.

Require verification of hot list data:

Law enforcement agencies should enact policies that require both independent verification of the information yielded from a hot list and real-time updating of hot list data. These steps would help prevent erroneous and potentially dangerous stops based on incorrect or outdated information.

Require transparency and invite community input into ALPR use and

policies: The public should have an opportunity to offer input into whether and how ALPRs are deployed. Given that a very small percentage of license plates scanned will actually be connected to a crime, communities may decide that money spent on ALPRs is better spent on other public safety needs, or that the privacy trade-offs are not worthwhile. If ALPRs are purchased, mechanisms must be in place to solicit feedback from interested community members on the policies governing their use. This public consultation process should also inform the types of crimes that merit inclusion on an ALPR hot list. Draft and final policies should be easily available to the public and must include the results of ongoing audits to detect and deter misuse of the system. ALPRs should not be deployed absent clear and enforceable use policies. Except in emergency circumstances, historical ALPR searches should not be conducted absent a warrant supported by probable cause.

Maintain audit logs: Law enforcement use of ALPR data should be logged and stored in a format that permits auditing. First, a log should maintain details about automated ALPR alerts, including the reason for the alert, whether any information was automatically shared with other agencies, and the outcome of the alert. Second, logs should track every time an officer seeks to access historical ALPR data as part of an investigation. This log should specify the officer and the crime being investigated and require evidence that a warrant has been obtained or specification of the exigent circumstances that mandate quicker access. If this functionality is not available through vendor platforms, law enforcement should establish internal access controls to ensure the same outcome.

Separately, each police department should establish a log that tracks and catalogs all the ways they receive, store, and share ALPR data. This includes the license plate reads collected

by their own devices, as well as those provided by other law enforcement agencies, by private vendors, and voluntarily by businesses and individuals. Many vendor platforms provide automated methods for tracking and updating authorized data flows, but each department should appoint an appropriate office to lead their efforts to track and maintain this log. When a department elects to share ALPR data with another law enforcement agency, the parties should enter into data sharing arrangements ensuring that policies regarding access control and retention are at least as strict as those of the originating agency. The receiving agency should also commit to entering into similar data sharing agreements for any downstream data sharing. Without adequate steps to protect downstream data sharing, even the most rigid policies will be insufficient once data is shared with a department that does not maintain the same level of protection.

Conduct audits for disparate im-

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should be periodically audited in order to protect against disparate impact on historically marginalized communities and constitutionally protected activities. These audits should evaluate the times and locations where ALPRs are used to ensure that they are not being used to disproportionately target particular communities or constitutionally protected activities such as protests. To facilitate this process, law enforcement agencies must keep records that detail the locations where ALPRs are deployed and the areas where historical searches are being run. Audits should also assess the types of investigations that merit a vehicle's inclusion on a hot list to ensure that low-level offenses are not effectively being used to target vulnerable communities. Audits should evaluate the extent to which ALPR data is used with other surveillance technologies — such as predictive policing algorithms or inclusion in gang databases — in a manner that could disproportionately harm

historically marginalized groups or constitutionally protected activity.

Conduct audits to ensure effective

safeguards: Every ALPR policy should include regular audits to evaluate safeguard effectiveness. These audits should ensure that ALPR data is only available to employees with a need to access the data, that their access is promptly terminated when no longer necessary, and that ALPR searches are appropriately limited to specific law enforcement investigations.

Ongoing oversight of the use of ALPR data within law enforcement agencies is an essential safeguard to detect and prevent officers' misuse of the system.

Endnotes

- 1 See Megan Brennan, “83% of U.S. Adults Drive Frequently; Fewer Enjoy It a Lot,” *Gallup*, July 9, 2018, <https://news.gallup.com/poll/236813/adults-drive-frequently-fewer-enjoy-lot.aspx>.
- 2 The focus of this report is law enforcement use of ALPR. Although there is an important discussion and analysis to be had regarding non-law enforcement applications, such as ALPR use to help institute congestion pricing or to automate toll collection, they are outside the scope of this report.
- 3 See, e.g., Katie Schoolov, “As Protests over the Killing of George Floyd Continue, Here’s How Police Use Powerful Surveillance Tech to Track Them,” *CNBC*, June 18, 2020, <https://www.cnbc.com/2020/06/18/heres-how-police-use-powerful-surveillance-tech-to-track-protestors.html>; Benjamin Wofford, “The Genius of Protesting in Car Caravans,” *Washingtonian*, June 1, 2020, <https://www.washingtonian.com/2020/06/01/the-genius-of-protesting-in-car-caravans/>; Caroline Haskins and Ryan Mac, “Here Are the Minneapolis Police’s Tools to Identify Protesters,” *BuzzFeed News*, May 29, 2020, <https://www.buzzfeednews.com/article/carolinehaskins1/george-floyd-protests-surveillance-technology>; and Catherine E. Shoichet, “They Can’t March in the Streets. So They’re Protesting in Their Cars Instead,” *CNN*, April 14, 2020, <https://www.cnn.com/2020/04/14/us/coronavirus-car-protests/index.html>.
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- 7** See, e.g., Crump, *You Are Being Tracked*, 5; International Association of Chiefs of Police, *Privacy Impact Assessment Report for the Utilization of License Plate Readers*, Alexandria, VA, September 2009, 2, 24–26, <http://web.archive.org/web/20131024095529/http://www.theiacp.org/LinkClick.aspx?fileticket=N%2BE2wvY%2F1QU%3D&tabid=87>; and *New York v. Davila*, 901 N.Y.S.2d 787, 789 (2010) (describing use of a hot list).
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- 23** State law may provide additional protections that go further than the guarantees of the Fourth Amendment. See, e.g., Electronic Communications Privacy Act, Cal. Penal Code § 1546.
- 24** *Carpenter v. United States*, 138 S. Ct. 2206, 2213 (2018) (quoting *Camara v. Municipal Court of City and County of San Francisco*, 387 U.S. 523, 528 (1967)).
- 25** See *Olmstead v. United States*, 277 U.S. 438, 464–66 (1928).
- 26** *Katz v. United States*, 389 U.S. 347, 351 (1967).
- 27** See *Smith v. Maryland*, 442 U.S. 735, 740 (1979) (quoting *Katz*, 389 U.S. at 361).
- 28** *Carpenter*, 138 S. Ct. at 2214 (quoting *Boyd v. United States*, 116 U.S. 616, 630 (1886); and *United States v. Di Re*, 332 U.S. 581, 595 (1948)).
- 29** *California v. Carney*, 471 U.S. 386, 392 (1985); see also *New York v. Class*, 475 U.S. 106, 113 (1986) (“Automobiles are justifiably the subject of pervasive regulation by the State. Every operator of a motor vehicle must expect that the State, in enforcing its regulations, will intrude to some extent upon that operator’s privacy.”).

- 30** See *U.S. v. Knotts*, 460 U.S. 276, 281 (1983); see also *Cardwell v. Lewis*, 417 U.S. 583, 590 (1974) (plurality opinion); *Town of Woodworth*, 132 So.3d 422 (La. App. 2013); *State v. Davis*, 239 P.3d 1002 (Ore. App. 2010); *State v. Myrick*, 659 A.2d 976; *Davila*, 901 N.Y.S.2d 787; *U.S. v. Diaz-Castaneda*, 494 F.3d 1146 (9th Cir. 2007); *U.S. v. Ellison*, 462 F.3d 557 (6th Cir. 2006); *Olabisiomotosho v. City of Houston*, 185 F.3d 521, 529 (5th Cir. 1999); *United States v. Walraven*, 892 F.2d 972, 974 (10th Cir. 1989); and *United States v. Matthews*, 615 F.2d 1279 (10th Cir. 1980).
- 31** *Delaware v. Prouse*, 400 U.S. 648, 662 (1979).
- 32** See *Knotts*, 460 U.S. at 283–84.
- 33** *Carpenter*, 138 S. Ct. at 2214.
- 34** *Kyllo v. United States*, 533 U.S. 27, 35 (2001).
- 35** *Riley v. California*, 134 S. Ct. 2473 (2014); *United States v. Jones*, 132 S. Ct. 945 (2012); *Carpenter*, 138 S. Ct. 2206.
- 36** *Jones*, 132 S. Ct. at 956 (Sotomayor, J., concurring) (quoting *United States v. Cuevas-Perez*, 640 F.3d 272, 285 (7th Cir. 2011) (Flaum, J., concurring)).
- 37** *Carpenter*, 138 S. Ct. at 2223.
- 38** *Carpenter*, 138 S. Ct. at 2223.
- 39** *Carpenter*, 138 S. Ct. at 2218.
- 40** *Carpenter*, 138 S. Ct. at 2216.
- 41** *Carpenter*, 138 S. Ct. at 2216.
- 42** See, e.g., Auditor of the State of California, *Automated License Plate Readers*, 1 (finding that “99.9 percent of the 320 million images Los Angeles stores are for vehicles that were not on a hot list when the image was made”).
- 43** *Carpenter*, 138 S. Ct. at 2210.
- 44** *Carpenter*, 138 S. Ct. at 2218–19 (quoting *Kyllo*, 533 U.S. at 36).
- 45** *Carpenter*, 138 S. Ct. at 2219–20.

- 46** See *United States v. Miller*, 425 U.S. 435, 443 (1976) (“The Fourth Amendment does not prohibit the obtaining of information revealed to a third party and conveyed by him to Government authorities, even if the information is revealed on the assumption that it will be used only for a limited purpose and the confidence placed in the third party will not be betrayed”).
- 47** *Carpenter*, 138 S. Ct. at 2220.
- 48** *Carpenter*, 138 S. Ct. at 2210.
- 49** See Department of Homeland Security, “Privacy Impact Assessment for CBP License Plate Reader Technology,” Washington, DC, July 6, 2020, 8, <https://www.dhs.gov/publication/dhscbppia-049-cbp-license-plate-reader-technology> (“The only way to opt out of such surveillance is to avoid the impacted area, which may pose significant hardships and be generally unrealistic”).
- 50** *United States v. Yang*, No. 18–10341, 2020 WL 2110973, at *6–7 (9th Cir. May 4, 2020), <https://cdn.ca9.uscourts.gov/datastore/opinions/2020/05/04/18-10341.pdf>.
- 51** *Commonwealth v. McCarthy*, 484 Mass. 493, 494 (2020), <https://cases.justia.com/massachusetts/supreme-court/2020-sjc-12750.pdf?ts=1587124946>.
- 52** *McCarthy*, 484 Mass. at 495.
- 53** *McCarthy*, 484 Mass. at 502.
- 54** *McCarthy*, 484 Mass. at 506.
- 55** *McCarthy*, 484 Mass. at 508–9.
- 56** See *Carpenter*, 138 S. Ct. at 2210 (“At any rate, the rule the Court adopts ‘must take account of more sophisticated systems that are already in use or in development,’ . . . and the accuracy of CSLI is rapidly approaching GPS-level precision.” (quoting *Kyllo*, 533 U.S. at 36)).
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- 62** *Green v. City and County of San Francisco*, 751 F.3d 1039, 1045 (9th Cir. 2014) (ruling that “an unconfirmed hit on the ALPR does not, alone, form the reasonable suspicion necessary to support an investigatory detention”).
- 63** *Green*, 751 F.3d 1039 at 1041.
- 64** *Green*, 751 F.3d 1039 at 1046.
- 65** *Green*, 751 F.3d 1039 at 1042.
- 66** *Green*, 751 F.3d 1039 at 1043.
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